

- Q.30 What are the assumption made in theory of Simple Bending?
- Q.31 Define overhanging beam and fixed beam.
- Q.32 Derive an expression for the strain energy stored in a body due to suddenly applied load.
- Q.33 Find the torque induced in a solid circular shaft of diameter 150 mm when the shaft transmits a power of 150KW at 180 rpm.
- Q.34 Explain Rankine Gourdan Formula.
- Q.35 Explain two types of Helical springs.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw a stress-strain graph for mule steel and explain Important points in the graph.
- Q.37 A solid shaft of 100 mm diameter transmit 160KW at 100 r.p.m. Find the maximum shear stress induced in the shaft.
- Q.38 With the help of neat diagram, explain different types of Beams.

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MSIL-120331/031731

1st Sem / Mech. Engg. (MSIL) Subject:- Strength of Materials

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the followings is a dimensionless quantity.
a) Shear stress b) Poisson's ratio
c) Compressive stress d) Modulus of rigidity
- Q.2 Any external force acting on a body is called
a) Stress b) Load
c) Strain d) None of these
- Q.3 The unit of strain is
a) N/cm^2 b) N/m
c) N/mm d) No Unit
- Q.4 Which of the following material is more elastic
a) Rubber b) Glass
c) Steel d) Wood
- Q.5 The Maximum strain energy stored by a body within the elastic limit is called
a) Resilience
b) Proof Resilience
c) Modulus of resilience
d) None of these

Q.6 The point of Contraflexure occurs in case of

- a) Cantilever beam
- b) Simply supported beam
- c) Over Hanging beam
- d) All of the above

Q.7 The unit of torque in S.I system is

- a) kgm b) kgcm
- c) Nm d) N/m

Q.8 The shaft are generally made of

- a) Mild steel b) Alloy steel
- c) Copper Alloy d) All of the above

Q.9 The shear stress is minimum at

- a) Axis of shaft b) Outer surface of shaft
- c) Inside of shaft d) None of the above

Q.10 The ratio of Load per unit deflection is called

- a) Strength b) Stiffness
- c) Brittleness d) Toughness

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define stress.

Q.12 Define slenderness Ratio.

Q.13 Define volumetric strain.

Q.14 Define modulus of elasticity.

Q.15 Define modulus of resilience.

Q.16 Write the SI unit of section modulus.

Q.17 Define cantilever beam.

Q.18 Define Torque.

Q.19 Define column.

Q.20 What is open coiled helical spring.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain strain with its types.

Q.22 Define elasticity and Toughness of materials.

Q.23 State and explain Hooke's law.

Q.24 What is Sagging Bending Moment and Hogging Bending moment?

Q.25 State and explain theorem of perpendicular axis.

Q.26 What are the assumption made in theory of pure torsion?

Q.27 Explain column with its types.

Q.28 Define spring and write its functions.

Q.29 A mild steel rod 30mm diameter is subjected to an axial pull of 50 KN. Determine stress induced in the rod.